Application Note/Product Note/Case Study

Applications for ultrafast Arbitrary Waveform Generators

Arbitrary waveform generators, or AWGs, are signal sources that can produce waveforms limited only by their sampling rate and bandwidth. Spectrum's new M5i.63xx series AWG family extends the frequency range of waveforms that PC based instruments can produce. The products feature sampling rates up to 10 Giga samples per second (GS/s) and output bandwidths up to 3.5 Gigahertz (GHz). They come packaged as either PCIe cards (M5i.63xx series) that plug directly into two PCIe slots of a suitable PC or as stand-alone NETBOX instruments (DN2.63x series) that connect directly to a PC or company network via Ethernet. The M5i.6357-x16 PCIe version is shown in Figure 1

Introduction

The AWGs are perfect for generating high-frequency signals up to their full bandwidth, with up to 100 picoseconds (ps) time resolution and an amplitude resolution of 16 bit. As such, they present test engineers, scientists, and designers with a cost-effective way to produce arbitrary waveforms up to microwave frequencies, with high purity and low distortion. The AWGs also feature massive onboard waveform memories of up to 8 Giga Samples per channel (16 GigaBytes), producing an 800 ms length at 10 GS/s. If necessary, the PCIe cards can produce even longer waveforms by streaming data directly over the bus from CPUs or GPUs at transfer rates up to 10 Gigabytes per second (10 GS/s). Designed for applications requiring automated and/or remote waveform generation, the AWGs are fully programmable using today's most popular programming languages. Spectrum's graphical user interface (GUI), SBench6, is also available for local interactive control, waveform creation, and display, and it includes various useful data import, export, and analysis tools. Let's look at some typical applications of these products.

Figure 1: The M5i.6357-x16 PCIe card with two differential channels and 10 GS/s output rate

Measuring RF Filter Characteristics

Measuring the frequency response of a filter requires a signal source and a measurement instrument like a digitizer, which should both have bandwidths greater than the filter bandwidth. Consider measuring the frequency response of a 50 MHz bandpass filter with a center frequency of 425 MHz. Possible broadband signal sources for frequency response testing include swept frequency sinewave, impulse functions, and white noise. The swept sinewave offers the highest dynamic range of measurements and is most commonly used.

The test requires a signal source to generate the input signal to the filter. In this example, an M5i.6350-x16 (10 GS/s, 2.5 GHz, 16 bit, one channel) is used to generate a linear sine sweep. A digitizer model M5i.3360-x16 (10 GS/s, 4.7 GHz, 12 bit, one channel) acquires the

© Spectrum Instrumentation GmbH, Germany

filter output, and the frequency response is measured. Each instrument is controlled, in this case, by Spectrum Instrumentation's SBench6 GUI.

SBench6 incorporates a function generator that can create complex waveforms using equations. The equation used in this example for a swept sinewave is shown in Figure 2.

The SBench6 function generator editor supports ten mathematical operators, eleven functions, four conditional functions, and two constants, which can be combined to create a broad range of waveforms.

The swept sinewave has to cover a frequency range greater than the filter's bandwidth. In this example, the sweep covered a 333 to 625 MHz range in 10 microseconds (µs). The signal described in the function generator is shown in Figure 3.

The entire sweep appears in the upper left grid. Horizontally expanded views of the beginning (bottom left grid) and the end (left center) show the change in its frequency. The waveform's Fast Fourier Transform (FFT), one of the calculations available in SBench6, shows the generated signal's frequency response (right-hand grid). Note that the frequency response is spectrally flat over the sweep range. Cursors mark the start and end frequencies, showing the nominal 333 to 625 MHz range annotated in the Info pane on the left. This signal is transferred to the AWG output as annotated in the Output Channel pane.

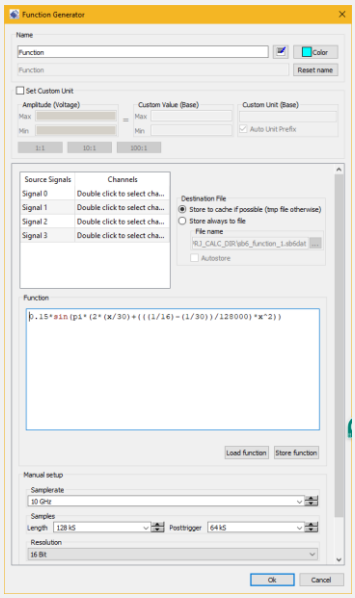

Figure 2: The SBench6 function generator dialog includes the complete description of the waveform, including its sample rate, duration, amplitude range, and equation.

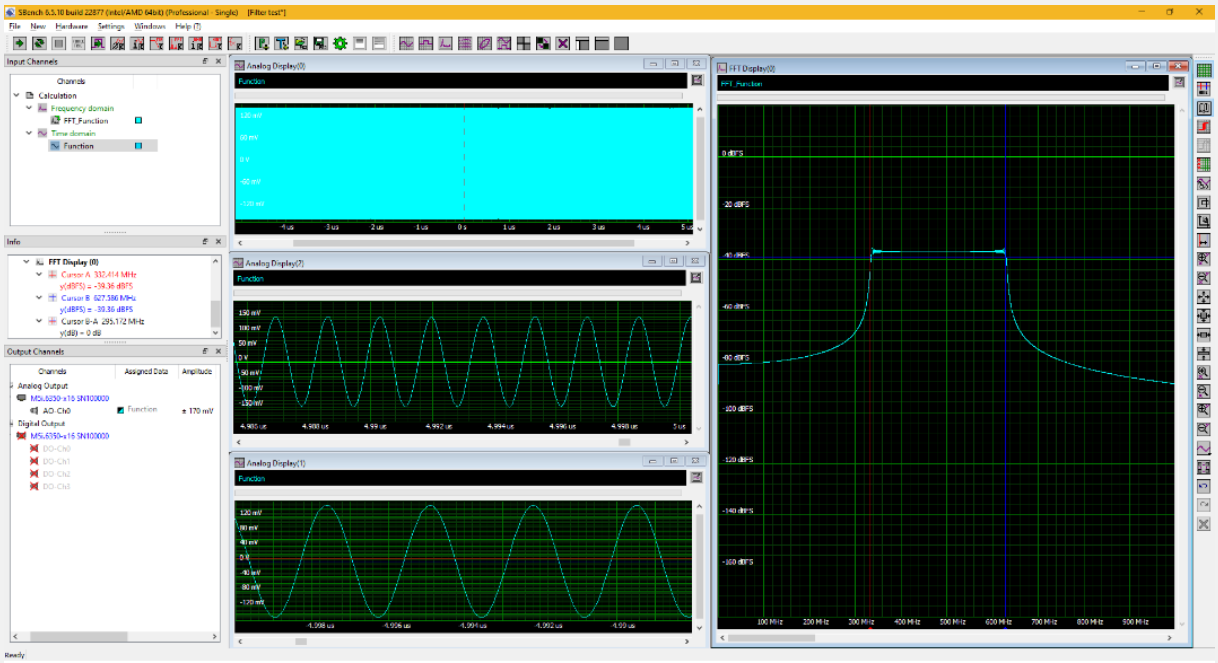

Figure 3. The signal created in the function generator with horizontally expanded views of the beginning and end and frequency response showing the range of frequencies output

The output of the AWG is applied to a properly terminated input of the filter under test. The output of the filter is connected to the M5i.3360-x16 digitizer. A separate instance of SBench6 controls the acquisition, displays the filter output signal, and performs the FFT to show the frequency response, as shown in Figure 4.

As shown in the M5i.63xx series, AWGs with 16-bit amplitude resolution can generate complex high-frequency signals like the linear frequency sweep shown with great precision and high signal purity.

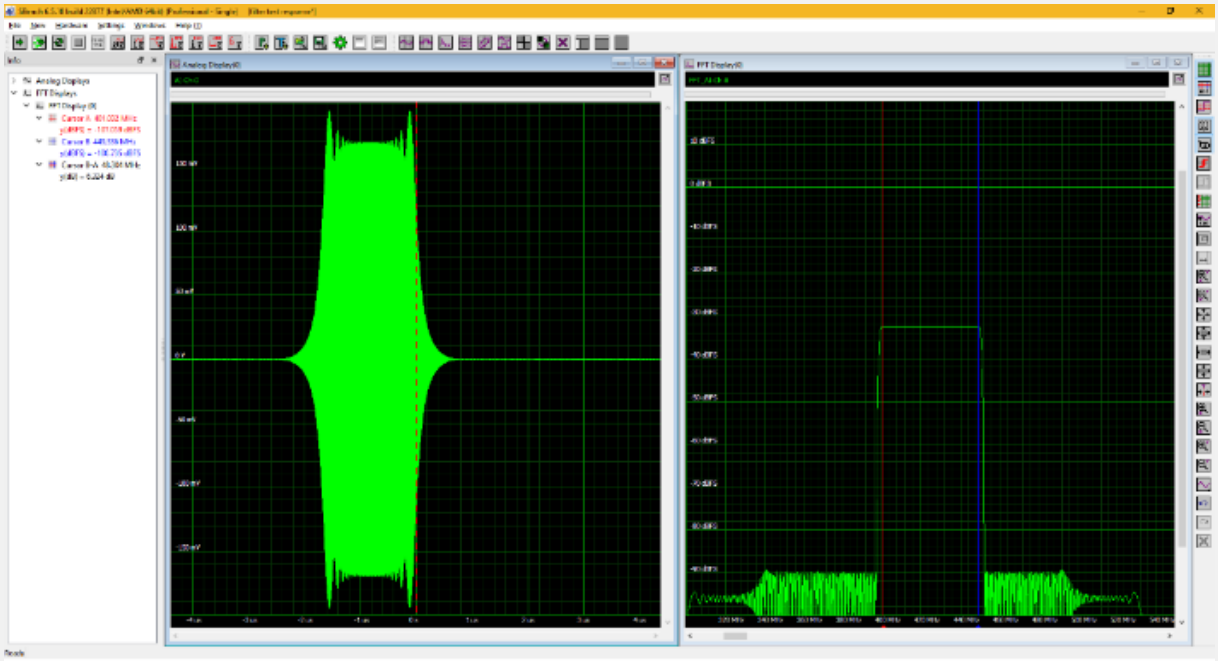

Figure 4: The digitizer acquired filter output (left grid) and frequency response (right grid). Cursors on the FFT trace read the nominal filter cutoff frequencies of 400 to 450 MHz, as shown in the info panel to the left.

Multi Tone Testing for Amplifier Linearity and EMC Measurements

Multitone testing is a common way of assessing the linearity of amplifiers. Tests are most commonly done using dual sinusoidal signals that have high signal purity. The two signals are applied to the input of an amplifier, and distortion products due to the non-linearity in the amplifier are measured. The Spectrum 63 series AWGs can generate the dual tone signal in a single channel with very low distortion, as shown in Figure 5.

© Spectrum Instrumentation GmbH, Germany

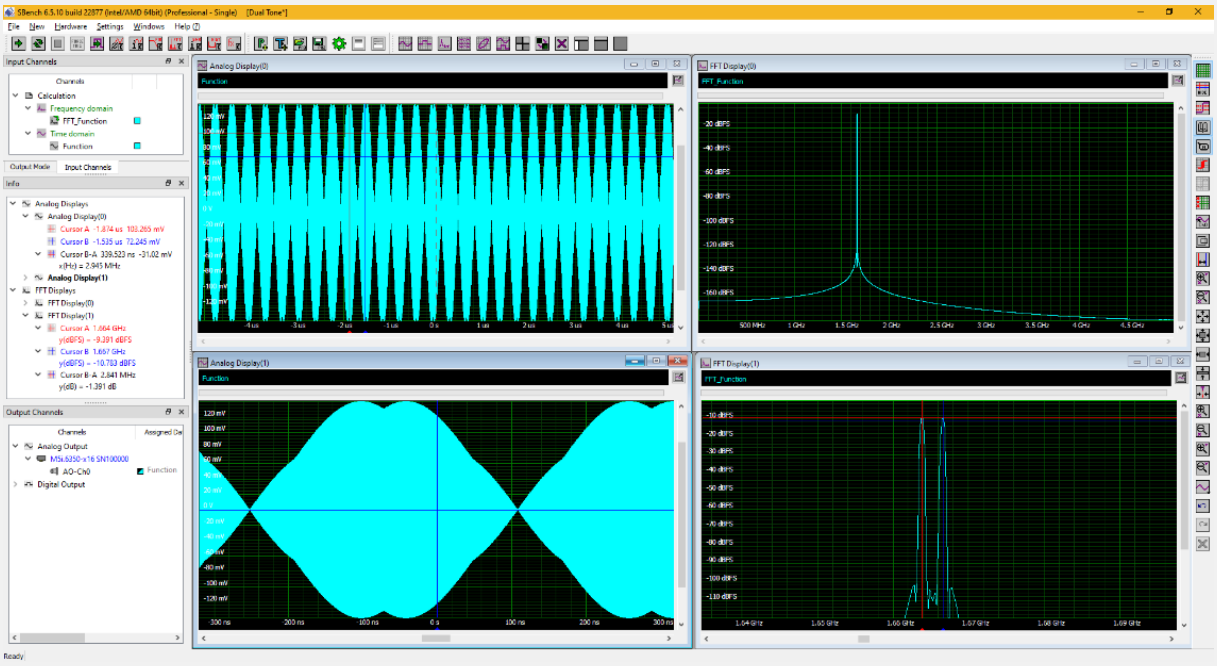

Figure 6: A dual-tone test signal consisting of the sum of two sine waves at 1.664 and 1.667 GHz produced by an M5i.6350-x16 AWG.

SBench6 creates the dual-tone waveform by summing two sine waves with frequencies of 1.6667 and 1.6664 GHz, which are created using a formula in its function generator. The resultant waveform shows a beat pattern equal to the frequency difference between the two sinewave components (top and bottom left grids). The FFT of the signal shows a

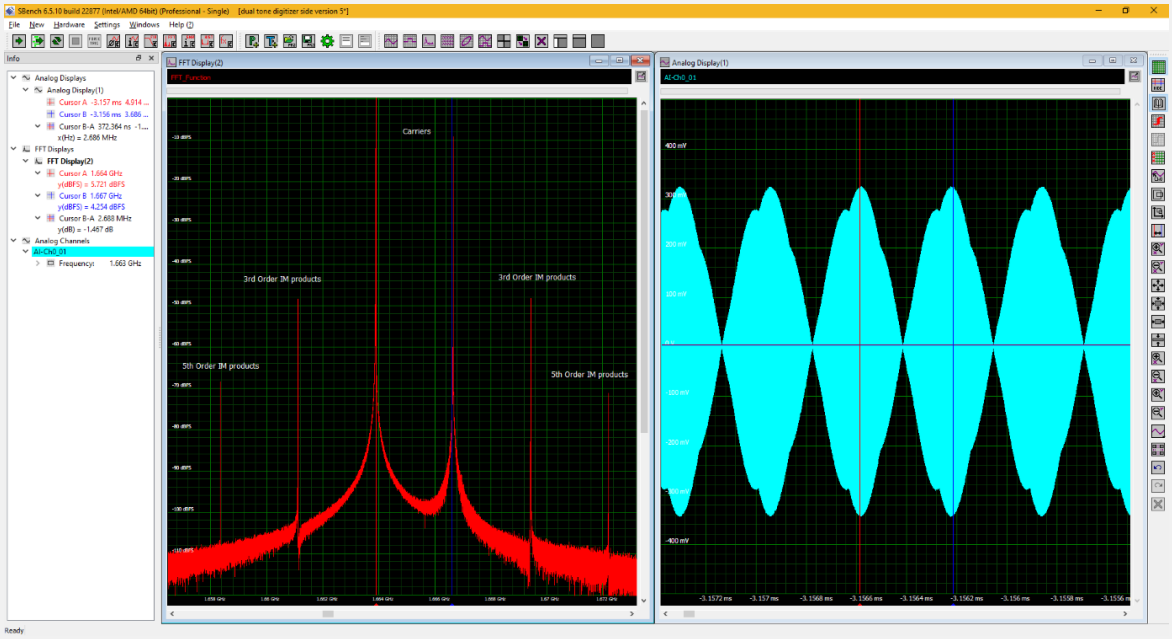

Figure 5: The results of a two-tone test show the effects of nonlinear operation on the signal. The output spectrum, expanded about the carrier frequencies, shows harmonic and mixing product signals due to device nonlinearity.

combined frequency peak at 1.66GHz (upper right). Horizontally expanding the FFT shows the two frequency components measured by cursors. This signal is coupled to the AWG output and applied to the amplifier under test. A Spectrum Instrumentation digitizer M5i.3360-x16 is used to measure the amplifier's output, shown in Figure 6.

The figure shows the acquired time signal on the right, which retains the beat frequency of the two carriers. The frequency spectrum of the acquired signal is expanded about the carriers to show the classic intermodulation mixing components. Cursors mark the carrier frequencies in the spectrum trace and the beat frequency in the time domain view. The third-order mixing products result from mixing one of the carrier signal's second harmonics with the other carrier. Other mixing products, such as the second and fifth order products, occur based on similar mixing of harmonics and the carrier signals. The amplitudes and frequencies of these harmonic and mixing components are used to compute several amplifier performance figures of merit. The purity of the signals from the source determines the limits of this type of measurement.

Multitone testing is also employed in electromagnetic compatibility (EMC) measurements. In this type of testing, multiple pairs of signals covering a broad band of frequencies are generated singly. Amplified signal pairs are emitted through an antenna to test a device's susceptibility to RF interference over a range of frequencies. The use of multitone signals reduces the test time for EMC qualification.

Modulated Signals

Radiofrequency communications signals are usually transmitted on modulated carriers. Modulation adds information to the carrier by varying its amplitude, frequency, or phase singly or simultaneously. The linear frequency sweep used for filter testing is a form of frequency modulation. Phase modulation is used in radar applications to improve radar

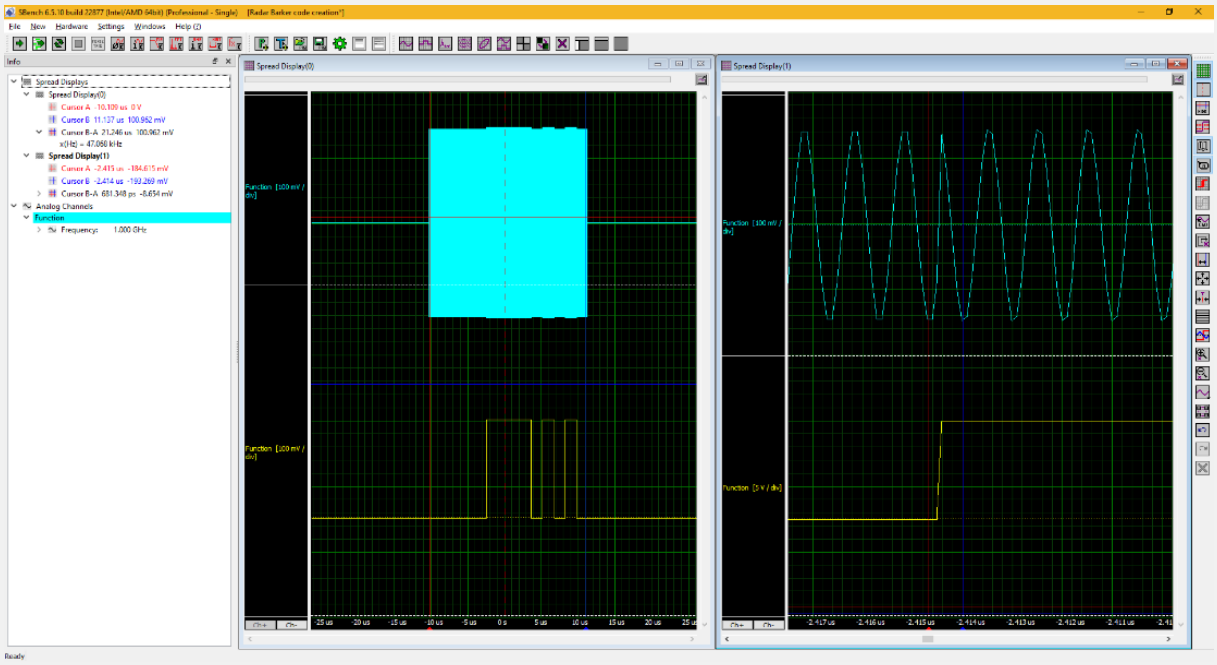

Figure 7: The results of a two-tone test show the effects of nonlinear operation on the signal. The output spectrum, expanded about the carrier frequencies, shows harmonic and mixing product signals due to device nonlinearity.

performance. Figure 7 shows the creation of a phase modulated radar pulse.

Application Note/Product Note/Case Study

The phase modulation technique used on a radar pulse breaks the pulse into segments, each transmitted with a specific phase shift. There are thirteen segments of equal length in this example. A code determines the selection of the phase shift. The common code is binary, where the code value switches between binary states, corresponding to a fixed phase shift controlled by the code sequence. The phase shift in this example is about 2 radians at each transition of the binary modulation signal. The waveform was created using SBench6, as shown in Figure 8.

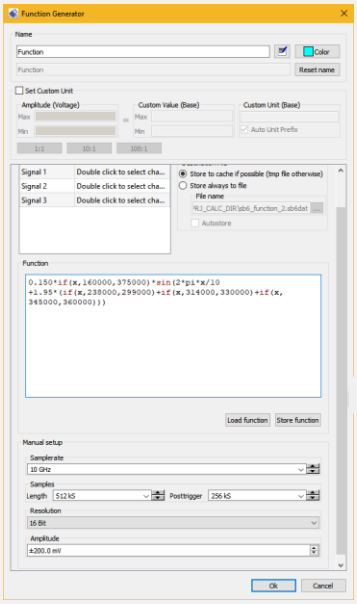

Figure 8: Creating a 1 GHz phase-modulated radar pulse with a duration of 21.5 µs using the binary data shown as the modulation source and an expanded view showing the phase shift at a binary state transition.

Importing Waveforms from a Digitizer

As utilitarian as the function software generator is, many situations require acquired waveforms from a digitizer, another instrument, or a third-party software package. This permits actual acquired signals to be duplicated for development and testing. SBench6 makes it simple to transfer these waveforms in various data formats.

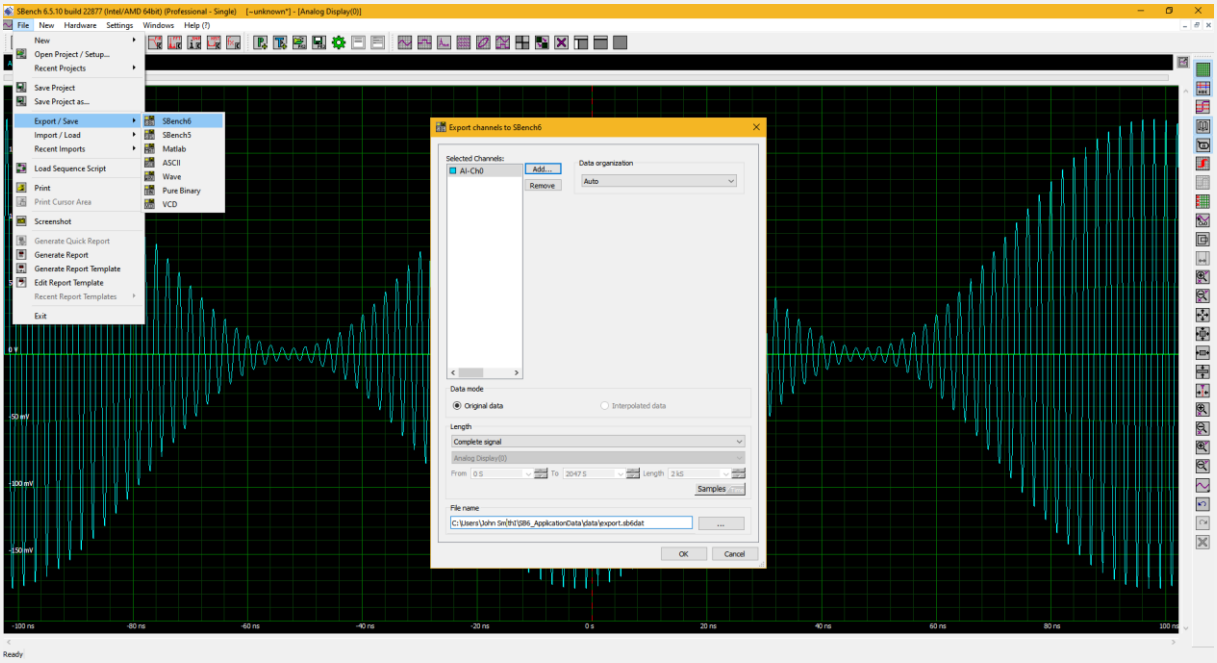

Figure 9: Using SBench6, waveforms can be exported from the digitizer to the host computer.

Transferring a waveform from a Spectrum Instrumentation digitizer to a 63 series AWG is accomplished by exporting acquired data from the digitizer to the host computer, as shown in Figure 9.

Waveforms acquired in the digitizer are exported using the Export/Save item in the File menu. The waveform can be exported in any of the seven data formats shown. The SBench6 format is used for transfers to the AWG. After selecting the SBench6 data format, the “Export channels to SBench6” dialog appears, allowing the user to select the desired waveform, or selected parts of it, to a selected file. Pressing the OK button transfers the waveform file to the computer.

Using a separate instance of SBench6, the AWG can import the file from the computer. Figure 10 outlines the process.

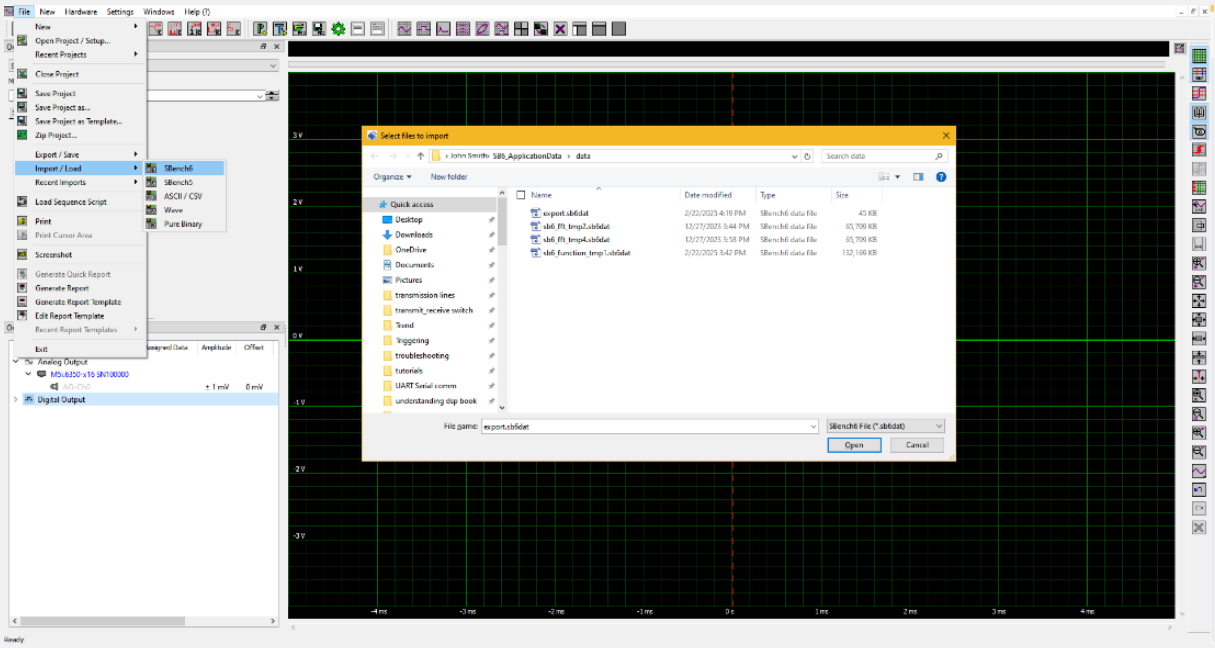

Figure 11: The sequence of operations, from left to right, to import the waveform from the host computer into the AWG.

The import selection under the file menu brings up the file format choices. Select the SBench6 format and navigate the location of the stored waveform. Pressing the Open button will import the waveform. The waveform will appear in the input channels panel under Import, as shown in Figure 11.

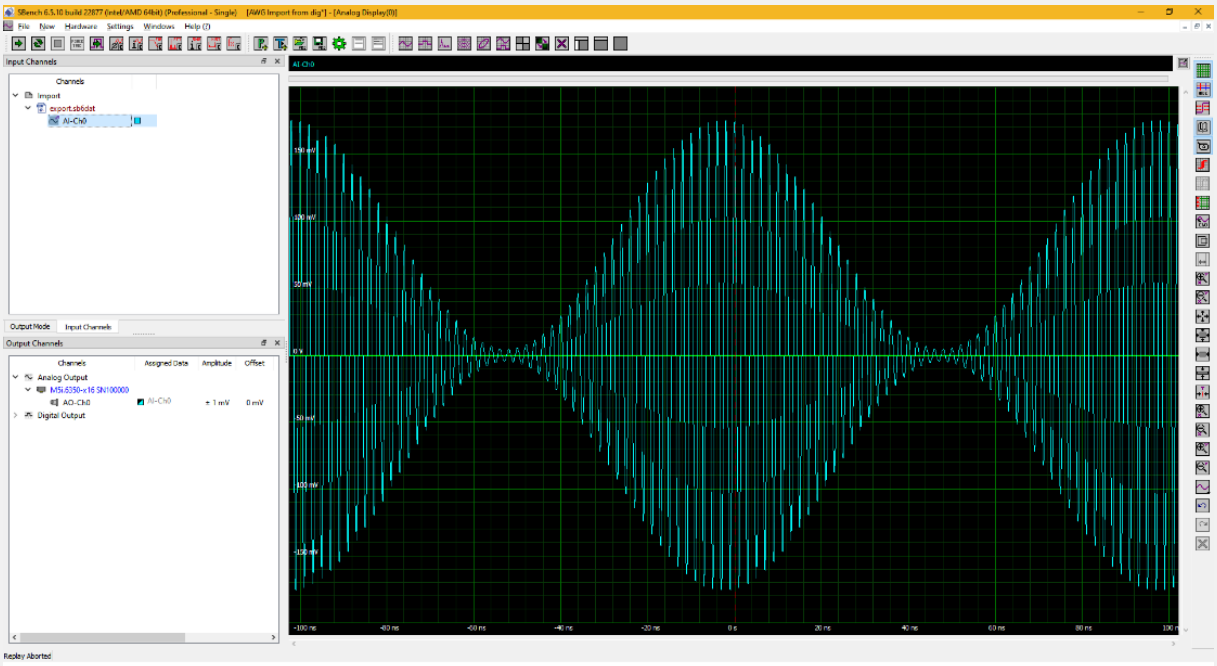

Figure 10: The imported waveform, displayed in an Analog Display grid.

The imported waveform can be displayed on an analog display grid and moved to an output

channel.

SBench6 is not your only choice for creating or transferring waveforms. Running under Windows or Linux, the AWGs are fully programmable using today's most popular and powerful software languages. AWGs and digitizers are shipped with software development kits (SDKs) for C++, C#, Python, VB.NET, Julia, and Java. Drivers are also provided for third-party software products LabVIEW and MATLAB. These tools can create and transfer waveforms but require specific programming.

Importing From Another Type of Instrument or Software Application

The AWG can also import waveforms from other types of instruments like digital oscilloscopes or third-party software applications. SBench6 can export MATLAB files directly, while for other sources ASCII files are used. As in the previous case, the SBench6 import function using the File pulldown is used. In this instance, the ASCII/CSV file type is selected. After selecting a target file, the Basic File Setup dialog appears in Figure 12.

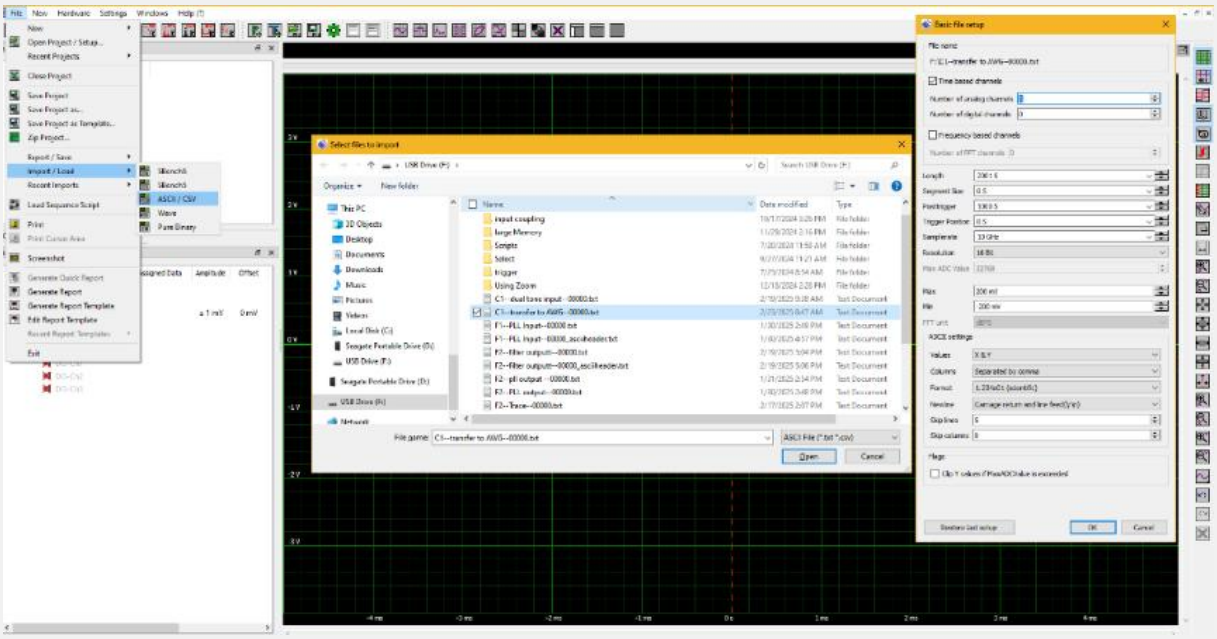

Figure 12: The sequence of operations, from left to right, to import an ASCII-encoded waveform file into the AWG.

The basic file setup describes the characteristics of the waveform contained in the ASCII file, including its length, the type of data values, numeric format, delineator character, sample rate, resolution, amplitude range, and the number of header lines to skip. Some entries, like the file length, are automatically derived from the file. The others must be entered to match the waveform file's format. Pressing OK will import the file into SBench6, as shown in Figure 13.

Figure 13: The imported waveform is listed in the Input Channels panel and is viewed in an analog display grid.

Once imported, the waveform is like any other SBench6 waveform and can be transferred to the output channel, measured, or processed.

Multiple Channel Operation

The Spectrum 63 series AWGs are available as PCIe cards and NETBOX instruments in single and dual-channel versions. For multi-input/multi-output (MIMO) applications requiring more channels, it's possible to connect multiple PCIe cards together using Spectrum's proprietary Star-Hub clock and trigger synchronization module. Star-Hub is an additional unit that can be installed on one card to allow phase stable synchronization of up to 8 AWG cards in a single system. Each card then shares a common clock and trigger, making it possible to create systems with up to sixteen channels clocked at 5 GS/s or eight channels up to 10 GS/s. Clock skew between single AWGs can be compensated in the range of ±127 ps.

Conclusion

The Spectrum 63 series of AWGs can generate almost any waveform with up to 10 GS/s sample rate, up to 3.5 GHz analog bandwidth, and 16-bit amplitude resolution, offering a perfect solution for almost every application. They are a way to generate high frequency, high purity, and low distortion waveforms using cost-effective off-the-shelf modular instruments.

© Spectrum Instrumentation GmbH, Germany